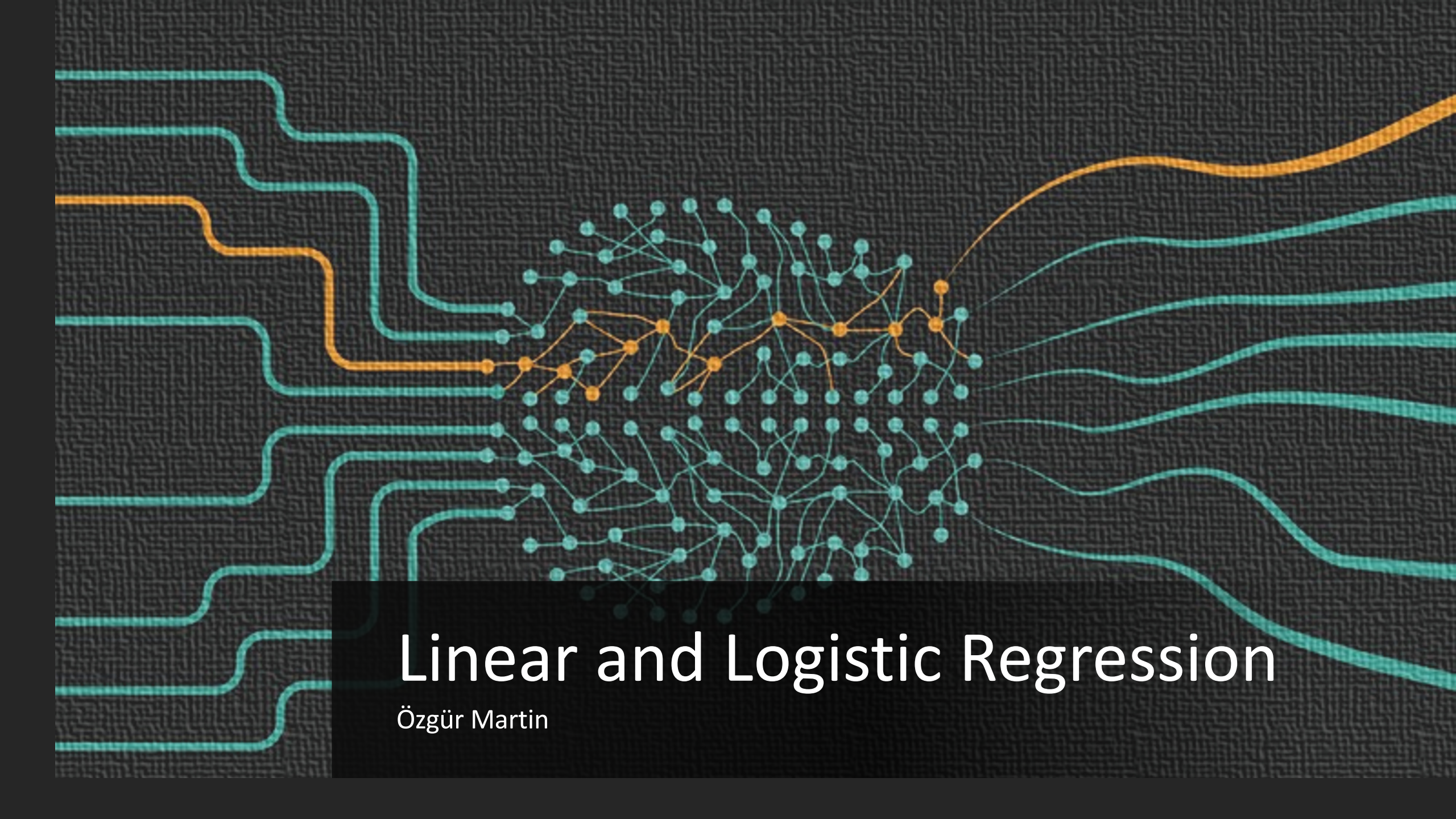
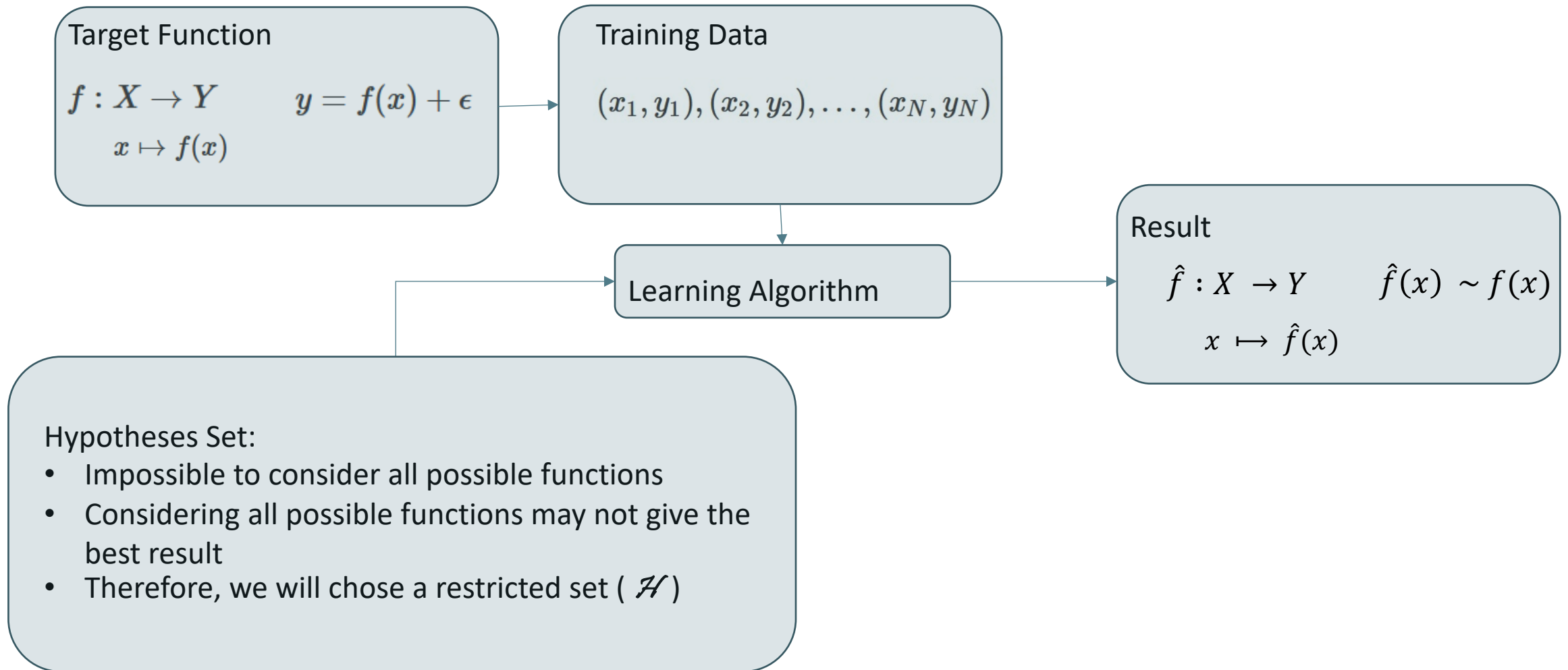


The background of the slide features a dark, textured surface. On the left, several thick, teal-colored lines enter from the edge and branch out in a stepped, circuit-like pattern. In the center, there is a dense network of small, light-blue circular nodes connected by thin, light-blue lines. A specific path of these nodes is highlighted with thicker orange lines, starting from the left and moving towards the right. On the right side, several thick, wavy teal lines exit the frame, and a single, thick orange line curves upwards from the network towards the top right corner.

Linear and Logistic Regression

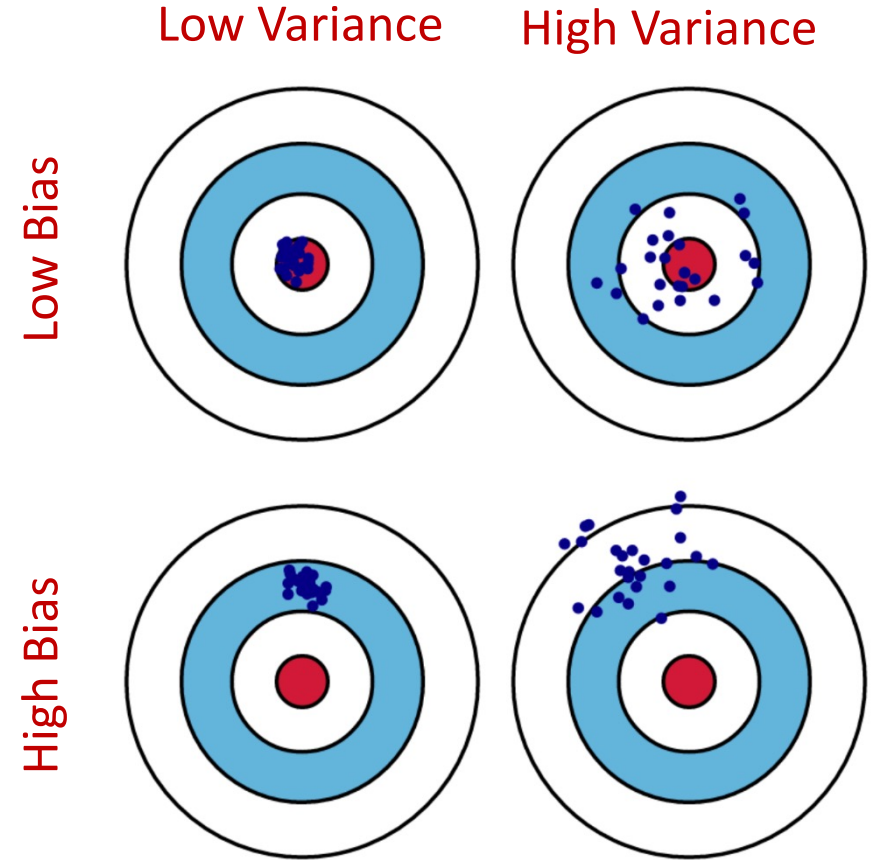
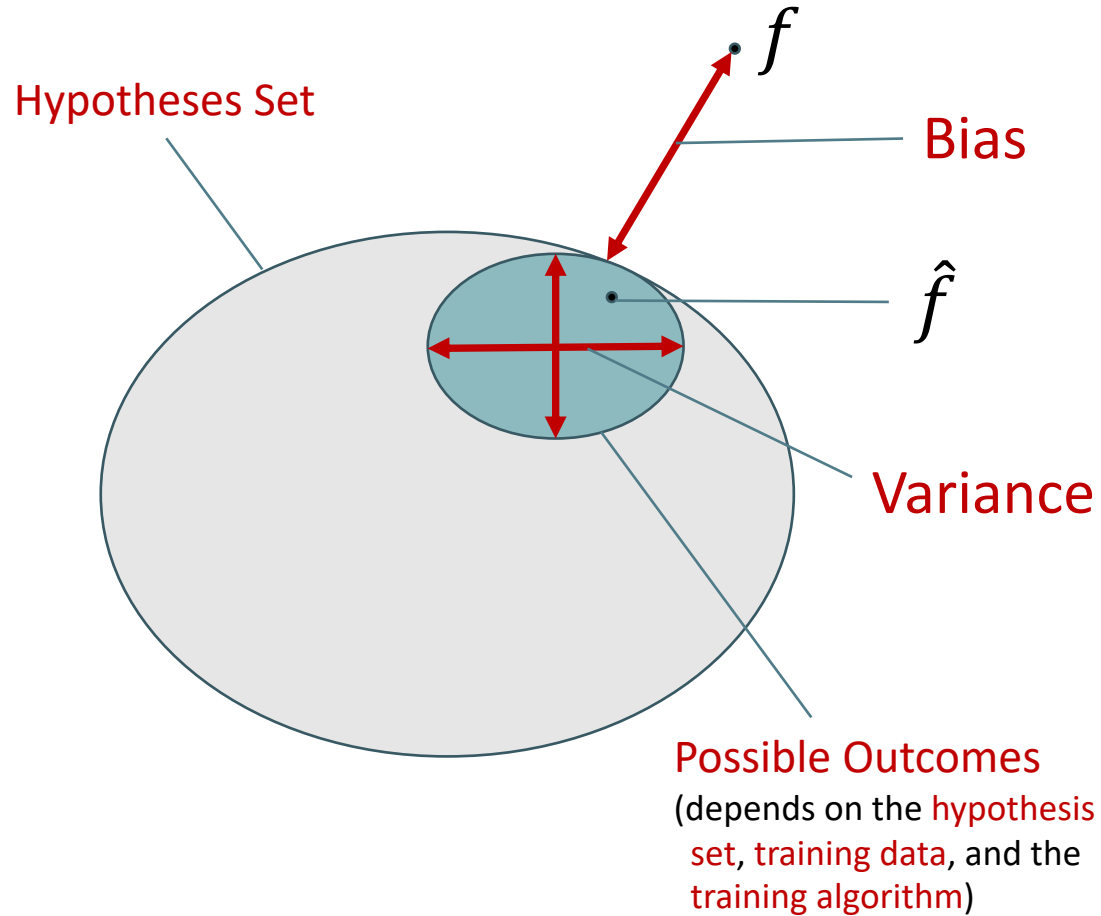
Özgür Martin

Recall: The Problem of Learning



Recall:

The Problem of Learning



Linear Regression

A simple approach for supervised learning

advertising.xlsx

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9
8.7	48.9	75	7.2
57.5	32.8	23.5	11.8
120.2	19.6	11.6	13.2

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we estimate the effect of each medium on sales?
- How accurately can we predict future sales?
- Is the relationship linear?

Simple Linear Regression

Recall:

$$Y = f(X) + \epsilon \xrightarrow{\text{Approx?}} \hat{Y} = \hat{f}(X)$$

$$Y \approx \hat{f}(X)$$

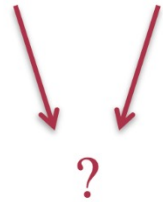
Simple Regression

$$Y \approx \beta_0 + \beta_1 X$$

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}.$$

$$Y \approx \beta_0 + \beta_1 X \quad (\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}.)$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$



Training Data

$$\{(x_i, y_i) : 1, \dots, n\}$$

$$y_i \approx \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i}, \quad i = 1, \dots, n$$

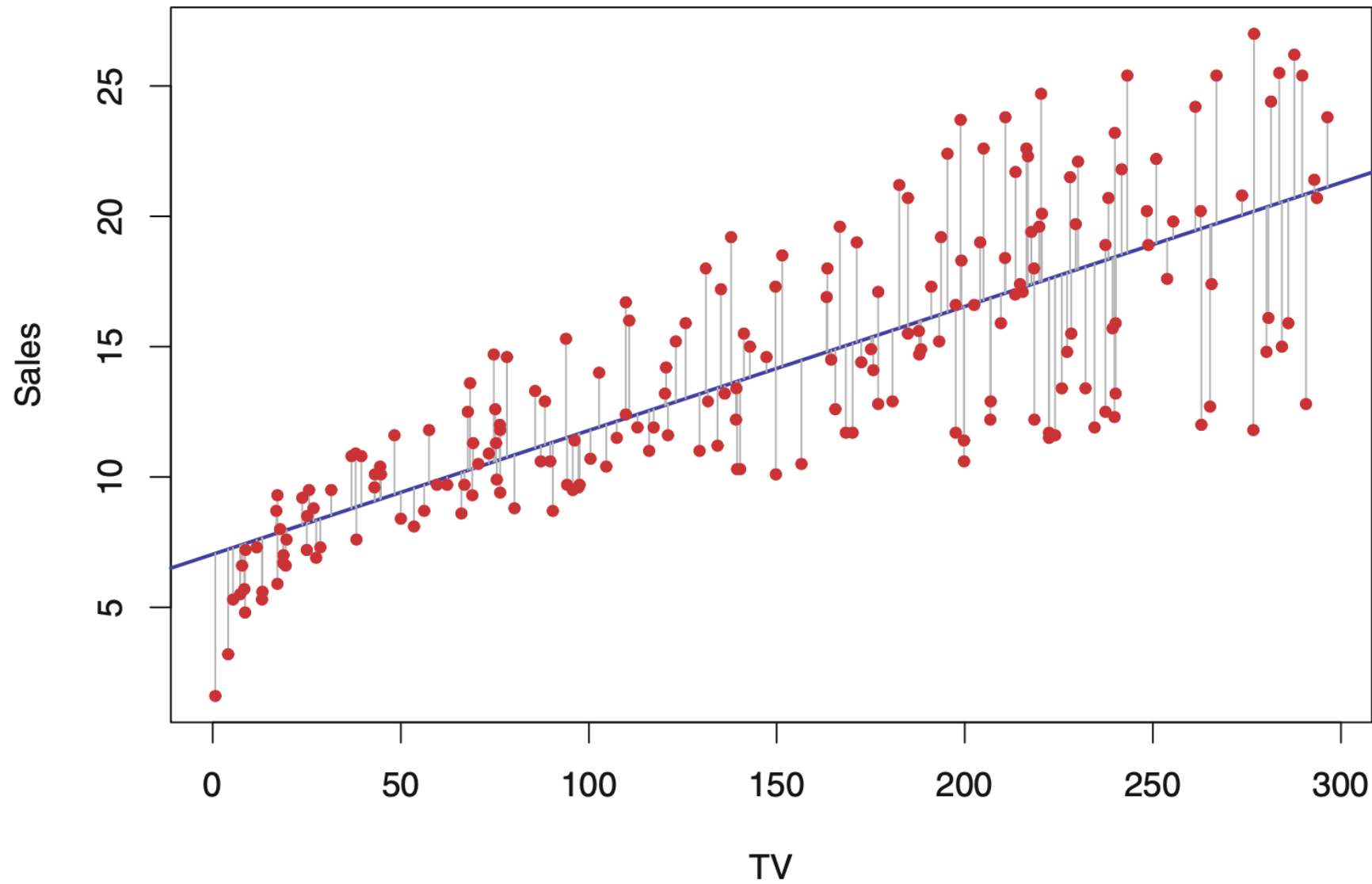
$$y_i \approx \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\hat{y}_i}, \quad i = 1, \dots, n$$

$$e_i = y_i - \hat{y}_i \quad (\textit{ith residual})$$

Residual Sum of Squares

$$\text{RSS} = e_1^2 + e_2^2 + \dots + e_n^2$$

(Aim: find $\hat{\beta}_0$ and $\hat{\beta}_1$ which minimize this sum)



Least Squares Method

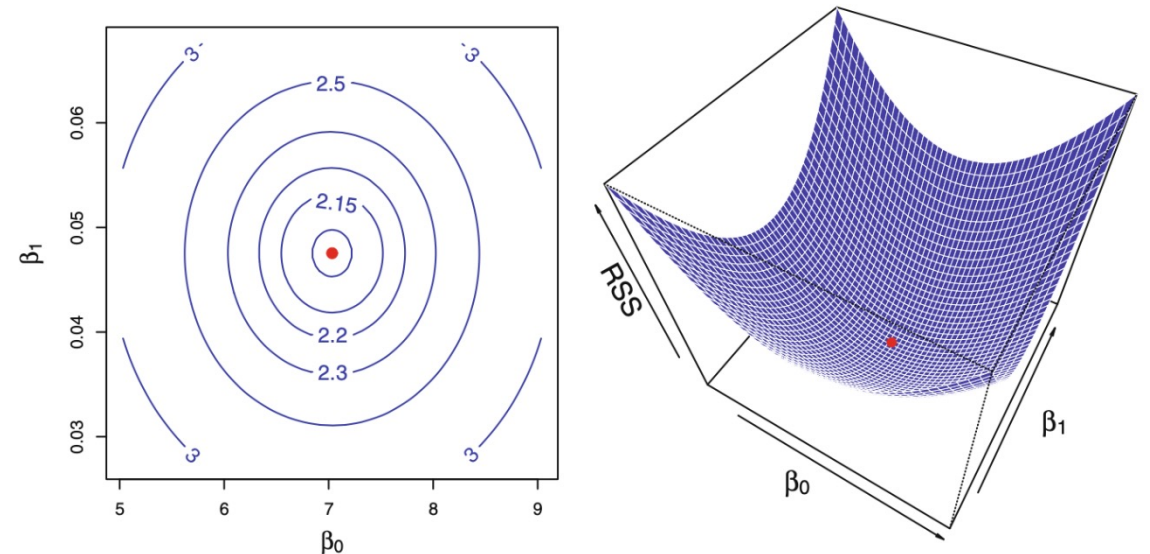
$$\min_{\hat{\beta}_0, \hat{\beta}_1} \text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \dots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

Solution

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$



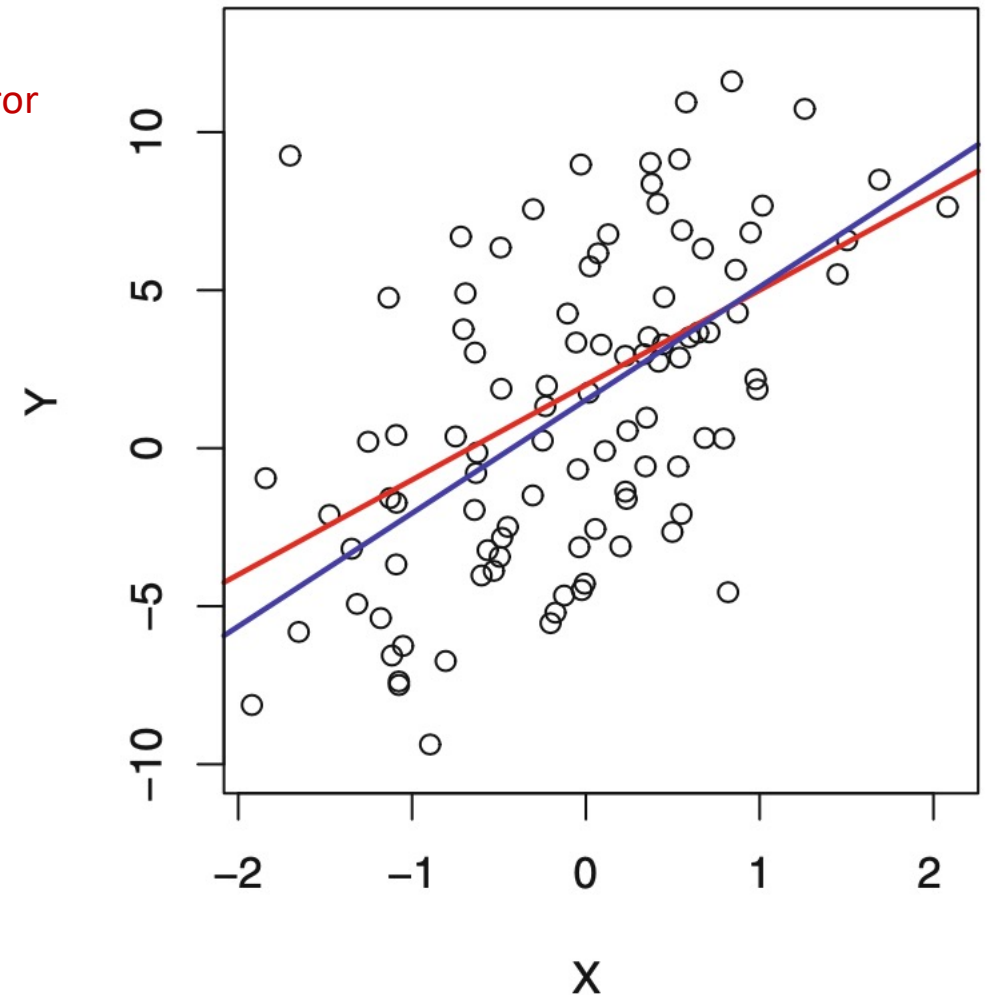
Simulated data from: $Y = 2 + 3X + \epsilon$ Random error

$\{(x_i, y_i): i = 1, \dots, n\}$ Simulated data set

Red line: The true relationship

$$f(X) = 2 + 3X$$

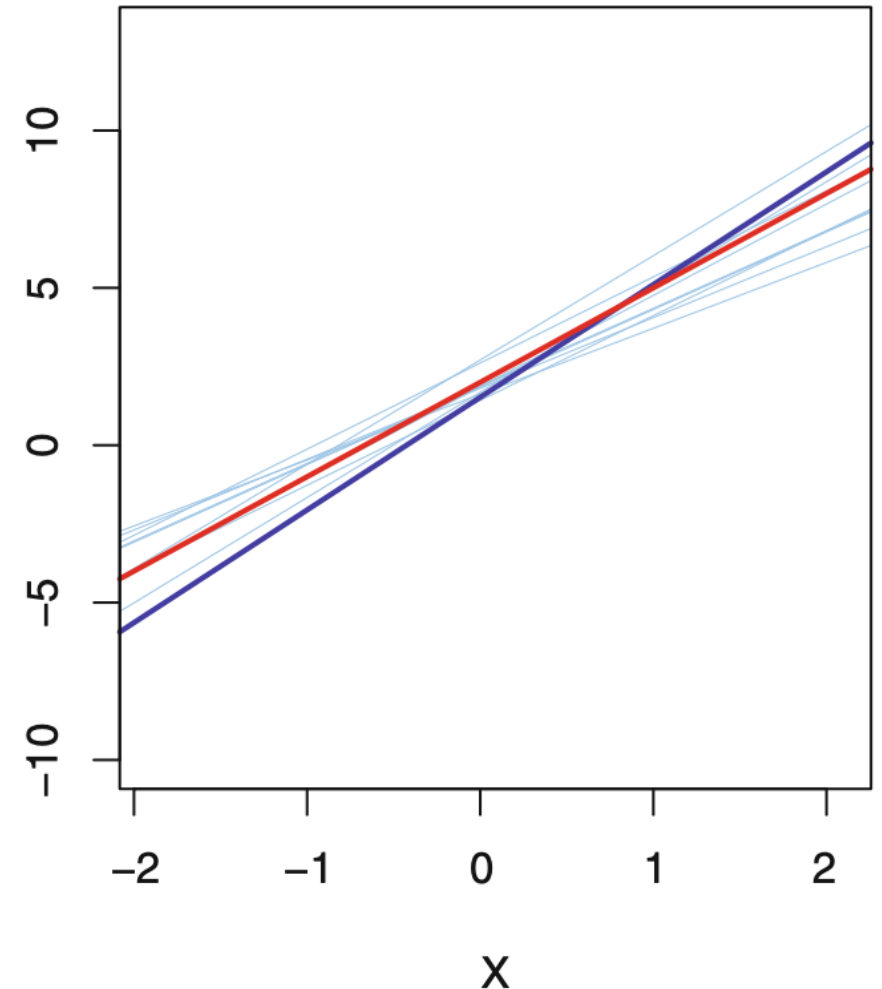
Blue line: The least squares estimate for $f(X)$



Let's create new simulated data sets and calculate the corresponding regression lines least squares method (LSM):

$$\begin{aligned} \{(x_i^{(1)}, y_i^{(1)}) : i = 1, \dots, n\} &\xRightarrow{\text{LSM}} \hat{\beta}_0^{(1)}, \hat{\beta}_1^{(1)} \\ \{(x_i^{(2)}, y_i^{(2)}) : i = 1, \dots, n\} &\xRightarrow{\text{LSM}} \hat{\beta}_0^{(2)}, \hat{\beta}_1^{(2)} \\ &\vdots \\ \{(x_i^{(N)}, y_i^{(N)}) : i = 1, \dots, n\} &\xRightarrow{\text{LSM}} \hat{\beta}_0^{(N)}, \hat{\beta}_1^{(N)} \end{aligned}$$

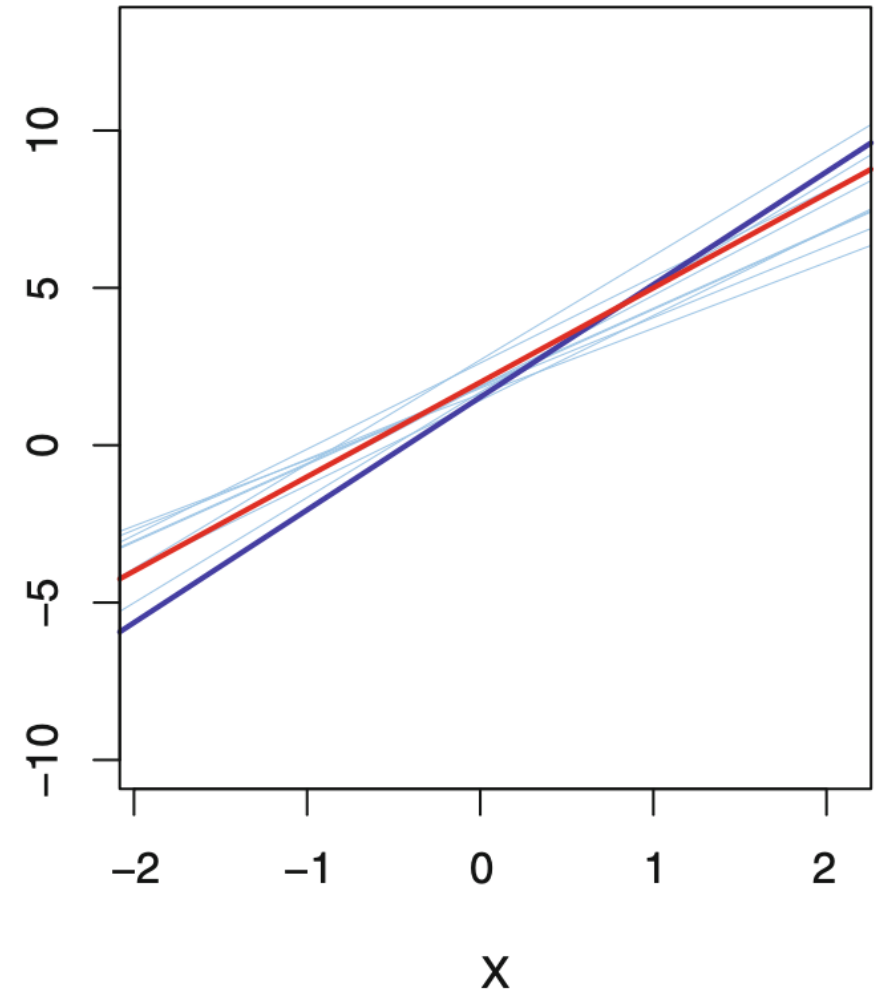
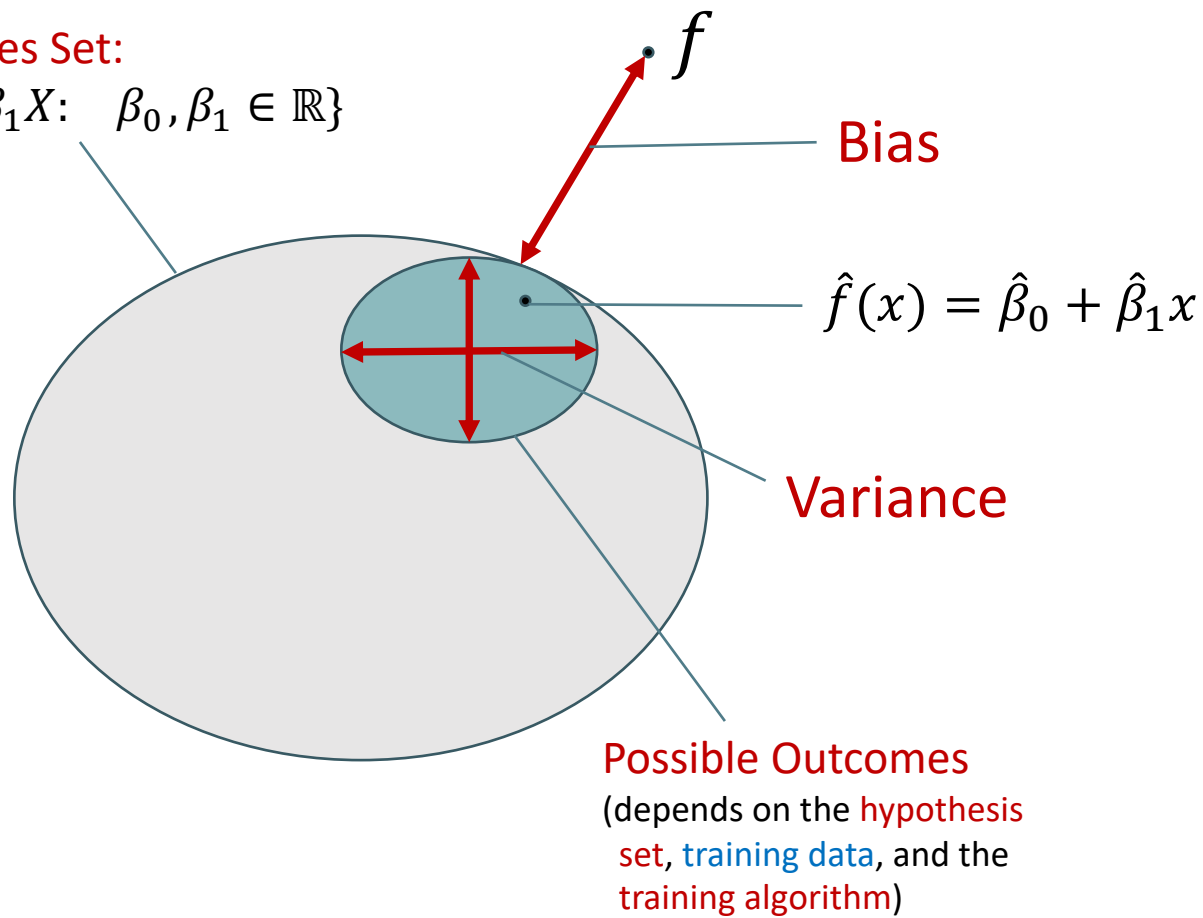
Observe the variation in the lines (models) as data changes!



Recall:

Hypotheses Set:

$$\{\beta_0 + \beta_1 X: \beta_0, \beta_1 \in \mathbb{R}\}$$



Simple Linear Regression Example

Advertising data: *advertising.xlsx*

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9
8.7	48.9	75	7.2
57.5	32.8	23.5	11.8
120.2	19.6	11.6	13.2

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}$$

Simple Linear Regression Example

```
import pandas as pd

df = pd.read_csv(r"C:\Users\
```

```
import statsmodels.api as sm

X = sm.add_constant(df.TV)
est = sm.OLS(df.Sales, X)
est2 = est.fit()
print(est2.summary())
```

```
=====
                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:          0.612
Model:                  OLS    Adj. R-squared:      0.610
Method:                 Least Squares    F-statistic:      312.1
Date:                  Tue, 12 Apr 2022    Prob (F-statistic):    1.47e-42
Time:                  16:07:10    Log-Likelihood:      -519.05
No. Observations:      200    AIC:              1042.
Df Residuals:          198    BIC:              1049.
Df Model:               1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	7.0326	0.458	15.360	0.000	6.130	7.935
TV	0.0475	0.003	17.668	0.000	0.042	0.053

```
=====
Omnibus:                0.531    Durbin-Watson:          1.935
Prob(Omnibus):           0.767    Jarque-Bera (JB):        0.669
Skew:                   -0.089    Prob(JB):                0.716
Kurtosis:                2.779    Cond. No.                 338.
=====
```

Linear Regression Model Assumptions

- The true relationship form is linear (Y is a linear function of X, plus random error)
- The error terms, ε_i are independent of the x values
- The error terms are random variables with mean 0 and constant variance, σ^2

(the constant variance property is called [homoscedasticity](#))

$$E[\varepsilon_i] = 0 \quad \text{and} \quad E[\varepsilon_i^2] = \sigma^2 \quad \text{for } (i = 1, \dots, n)$$

- The random error terms, ε_i , are not correlated with one another, so that

$$E[\varepsilon_i \varepsilon_j] = 0 \quad \text{for all } i \neq j$$

Expectation from Linear Regression

Prediction

$$x_0 = \begin{bmatrix} \circ \\ \vdots \\ \circ \end{bmatrix} \longrightarrow y_0 = ?$$

What?

YES

Inference

$$x_0 = \begin{bmatrix} \bullet \\ \vdots \\ \bullet \end{bmatrix} \xrightarrow{?} y_0$$

How?

YES

Multiple Linear Regression

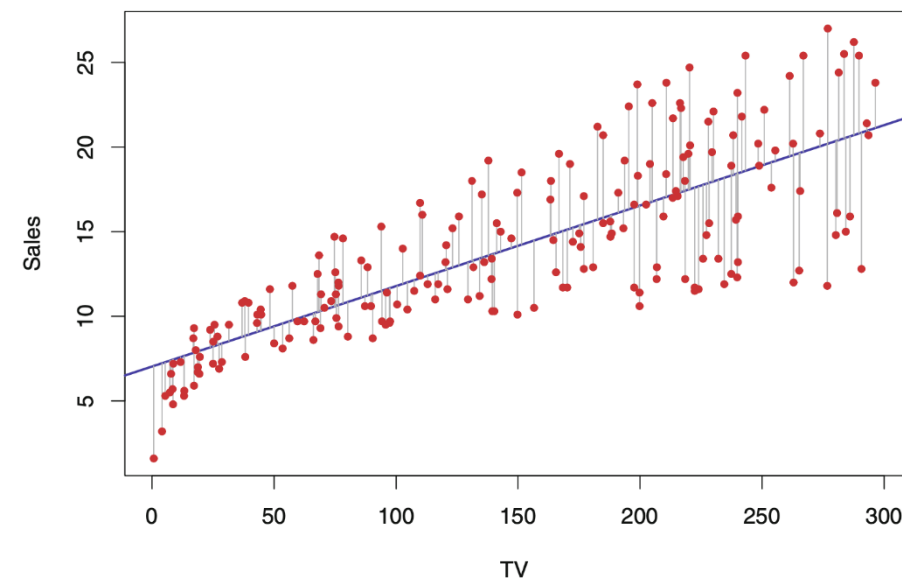
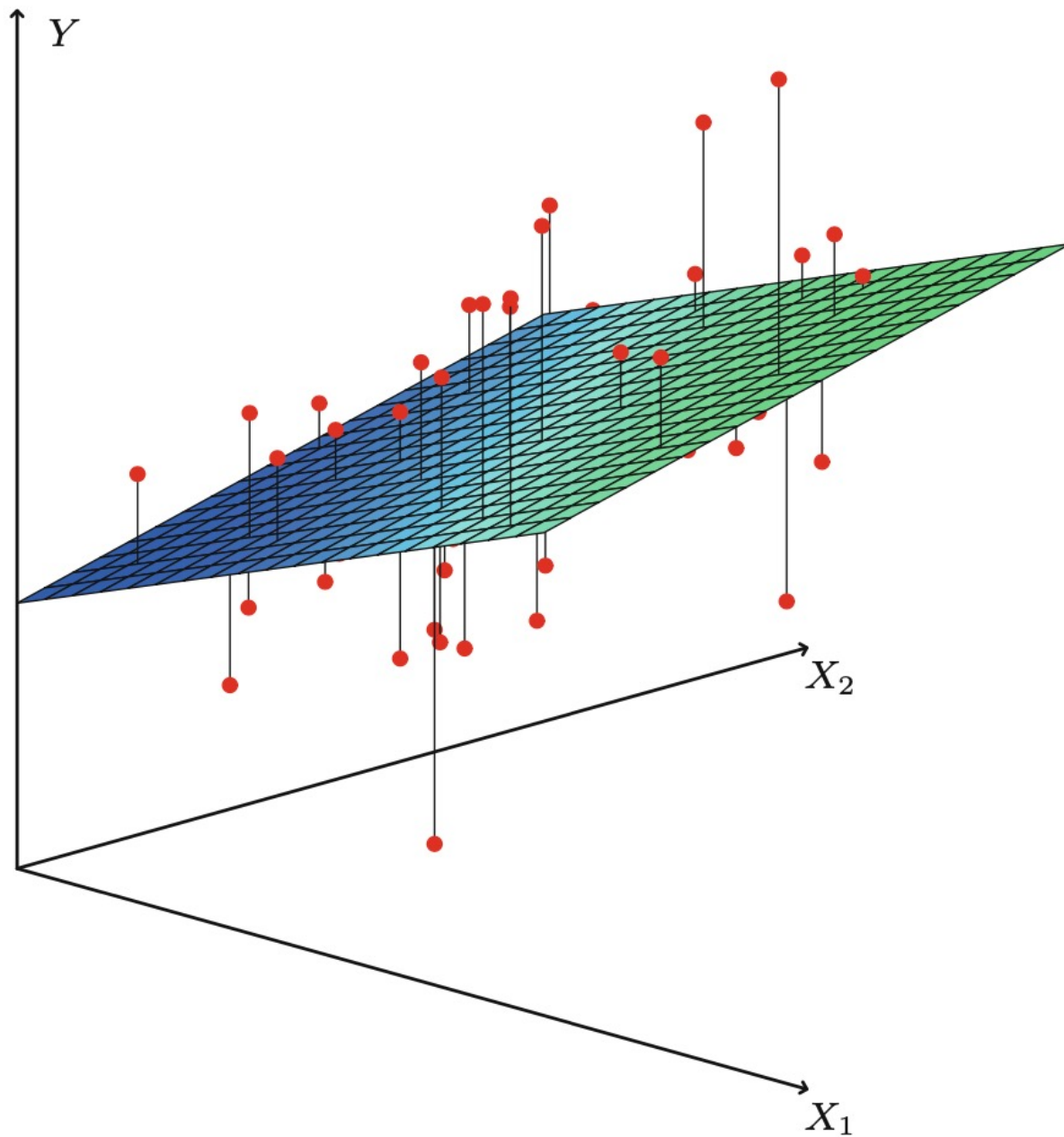
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon$$

Training Data: $\{(x_i, y_i) : 1, \dots, n\}$

$$y_i \approx \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_p x_{ip}}_{\hat{y}_i}, \quad i = 1, \dots, n$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



RSS

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$



Convex Function



$$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$$

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_1^\top \\ 1 & x_2^\top \\ \vdots & \vdots \\ 1 & x_n^\top \end{bmatrix}_{n \times (p+1)} \quad \mathbf{y}^\top = (y_1, \dots, y_n)$$

$$\boldsymbol{\beta}^\top = (\beta_0, \beta_1, \dots, \beta_p)$$

$$\hat{\boldsymbol{\beta}}_{\text{LS}} = \arg \min_{\boldsymbol{\beta}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

(with the assumption of full rank)

$$\hat{\boldsymbol{\beta}}_{\text{LS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

$$\hat{y}_0 = (1 \ x_0^\top) \hat{\boldsymbol{\beta}}_{\text{LS}}$$

Standard Multiple Regression Assumptions

- The values x_i and the error terms ε_i are independent
- The error terms are random variables with mean 0 and a constant variance, σ^2 .

$$E[\varepsilon_i] = 0 \quad \text{and} \quad E[\varepsilon_i^2] = \sigma^2 \quad \text{for } (i = 1, \dots, n)$$

(The constant variance property is called **homoscedasticity**)

- The random error terms, ε_i , are not correlated with one another, so that

$$E[\varepsilon_i \varepsilon_j] = 0 \quad \text{for all } i \neq j$$

- It is not possible to find a set of numbers, $c_0, c_1, \dots, c_k$, such that

$$c_0 + c_1 x_{1i} + c_2 x_{2i} + \dots + c_k x_{ki} = 0$$

(This is the property of no linear relation for the x_i 's)

Advertisement Example

```

                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                  0.897
Model:                          OLS     Adj. R-squared:             0.896
Method:                        Least Squares   F-statistic:                859.6
Date:                          Tue, 12 Apr 2022   Prob (F-statistic):        4.83e-98
Time:                          17:33:03    Log-Likelihood:            -386.20
No. Observations:                200      AIC:                       778.4
Df Residuals:                    197      BIC:                       788.3
Df Model:                        2
Covariance Type:                nonrobust
=====
                                coef    std err          t      P>|t|      [0.025    0.975]
-----
const                2.9211      0.294      9.919      0.000      2.340      3.502
TV                   0.0458      0.001     32.909      0.000      0.043      0.048
Radio               0.1880      0.008     23.382      0.000      0.172      0.204
=====
Omnibus:                60.022    Durbin-Watson:            2.081
Prob(Omnibus):          0.000    Jarque-Bera (JB):         148.679
Skew:                  -1.323    Prob(JB):                 5.19e-33
Kurtosis:               6.292    Cond. No.                  425.
=====
```

Advertisement Example

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.646
Model:                  OLS      Adj. R-squared:             0.642
Method:                 Least Squares    F-statistic:              179.6
Date:                  Tue, 12 Apr 2022    Prob (F-statistic):       3.95e-45
Time:                  17:33:20    Log-Likelihood:           -509.89
No. Observations:      200    AIC:                      1026.
Df Residuals:          197    BIC:                      1036.
Df Model:               2
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	5.7749	0.525	10.993	0.000	4.739	6.811
TV	0.0469	0.003	18.173	0.000	0.042	0.052
Newspaper	0.0442	0.010	4.346	0.000	0.024	0.064

```

=====
Omnibus:                0.658    Durbin-Watson:           1.969
Prob(Omnibus):          0.720    Jarque-Bera (JB):        0.415
Skew:                   -0.093    Prob(JB):                0.813
Kurtosis:               3.122    Cond. No.                 410.
=====

```

Advertisement Example

```

                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                  0.333
Model:                          OLS      Adj. R-squared:             0.326
Method:                        Least Squares  F-statistic:                49.11
Date:                          Tue, 12 Apr 2022  Prob (F-statistic):      4.96e-18
Time:                          17:33:35    Log-Likelihood:             -573.24
No. Observations:                200      AIC:                       1152.
Df Residuals:                    197      BIC:                       1162.
Df Model:                        2
Covariance Type:                nonrobust
=====
                                coef    std err          t      P>|t|      [0.025    0.975]
-----
const                9.1889      0.628     14.640     0.000      7.951     10.427
Radio                 0.1990      0.022      9.101     0.000      0.156      0.242
Newspaper             0.0066      0.015      0.446     0.656     -0.023      0.036
=====
Omnibus:                19.261    Durbin-Watson:              1.943
Prob(Omnibus):           0.000    Jarque-Bera (JB):           21.768
Skew:                   -0.765    Prob(JB):                   1.88e-05
Kurtosis:                3.519    Cond. No.                    92.0
=====
```

Advertisement Example

```
=====
                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.897
Model:                  OLS      Adj. R-squared:             0.896
Method:                 Least Squares    F-statistic:            570.3
Date:                  Tue, 12 Apr 2022    Prob (F-statistic):      1.58e-96
Time:                  17:36:13    Log-Likelihood:         -386.18
No. Observations:      200    AIC:                    780.4
Df Residuals:          196    BIC:                    793.6
Df Model:               3
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                2.9389        0.312        9.422      0.000        2.324        3.554
TV                   0.0458        0.001       32.809      0.000        0.043        0.049
Radio                0.1885        0.009       21.893      0.000        0.172        0.206
Newspaper           -0.0010        0.006       -0.177      0.860       -0.013        0.011
=====
Omnibus:              60.414    Durbin-Watson:           2.084
Prob(Omnibus):        0.000    Jarque-Bera (JB):        151.241
Skew:                 -1.327    Prob(JB):                1.44e-33
Kurtosis:              6.332    Cond. No.                 454.
=====
```

Advertisement Example

```
=====
                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.897
Model:                  OLS      Adj. R-squared:             0.896
Method:                 Least Squares    F-statistic:            570.3
Date:                  Tue, 12 Apr 2022    Prob (F-statistic):      1.58e-96
Time:                  17:36:13    Log-Likelihood:         -386.18
No. Observations:      200    AIC:                    780.4
Df Residuals:          196    BIC:                    793.6
Df Model:               3
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                2.9389        0.312        9.422      0.000        2.324        3.554
TV                   0.0458        0.001       32.809      0.000        0.043        0.049
Radio                0.1885        0.009       21.893      0.000        0.172        0.206
Newspaper           -0.0010        0.006       -0.177      0.860       -0.013        0.011
=====
Omnibus:              60.414    Durbin-Watson:           2.084
Prob(Omnibus):        0.000    Jarque-Bera (JB):        151.241
Skew:                 -1.327    Prob(JB):                1.44e-33
Kurtosis:              6.332    Cond. No.                 454.
=====
```

Advertisement Example: Correlations

	TV	Radio	Newspaper	Sales
TV	1.000000	0.054809	0.056648	0.782224
Radio	0.054809	1.000000	0.354104	0.576223
Newspaper	0.056648	0.354104	1.000000	0.228299
Sales	0.782224	0.576223	0.228299	1.000000

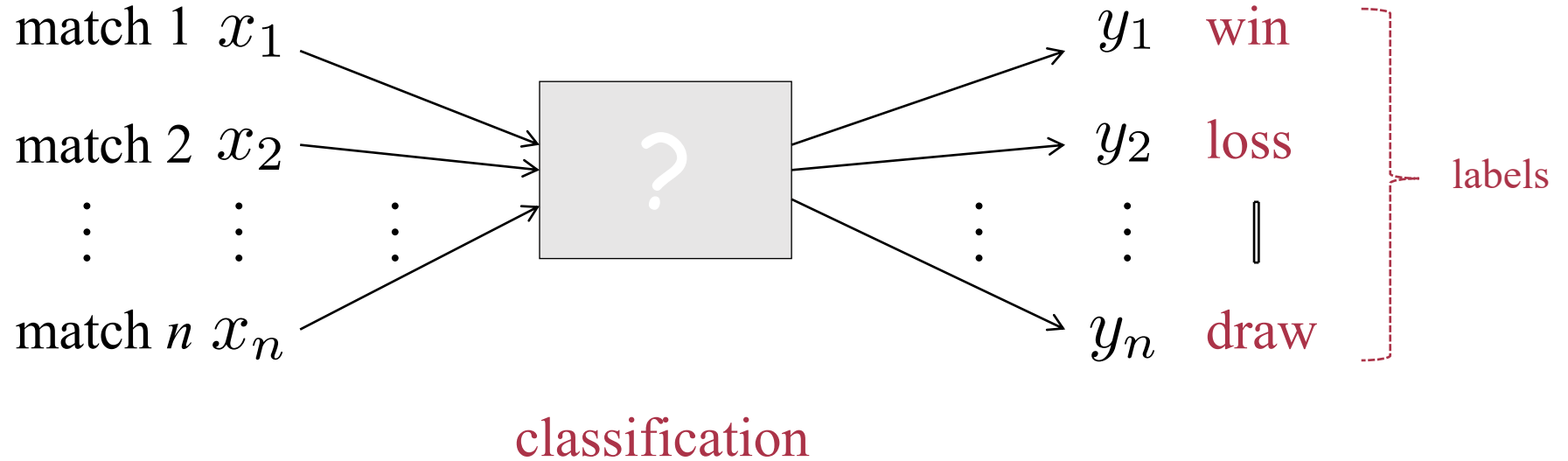
Polynomial Regression

Actually, it is just **Multiple Linear Regression**:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_n X^n + \epsilon,$$

where $X_1 = X, X_2 = X^2, \dots, X_n = X^n$.

Classification



training data
 $\{(x_i, y_i) : 1, \dots, n\}$

Logistic Regression

X_1 : balance

X_2 : income

X_3 : student (1), not student (0)

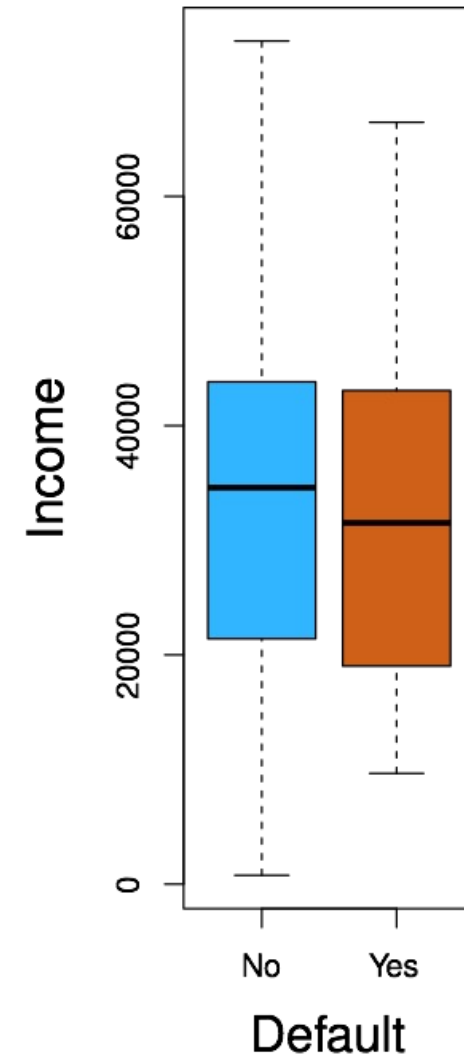
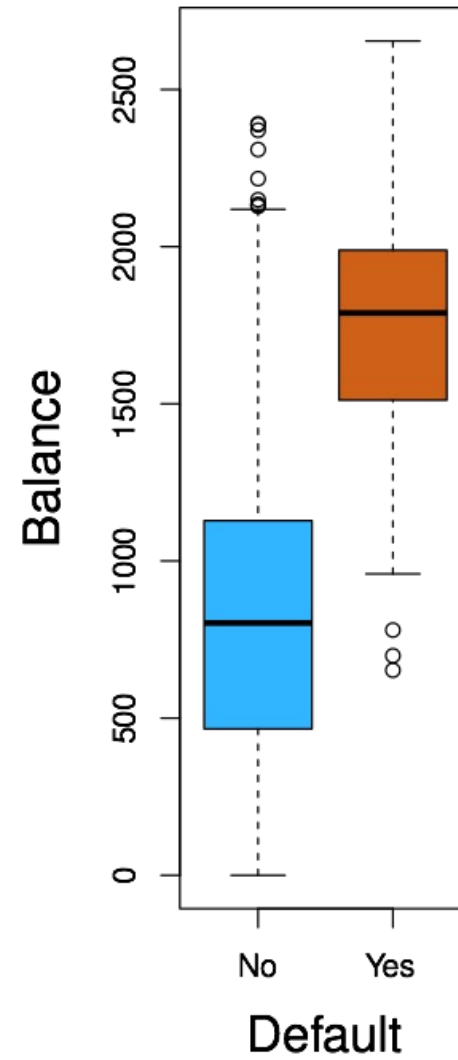
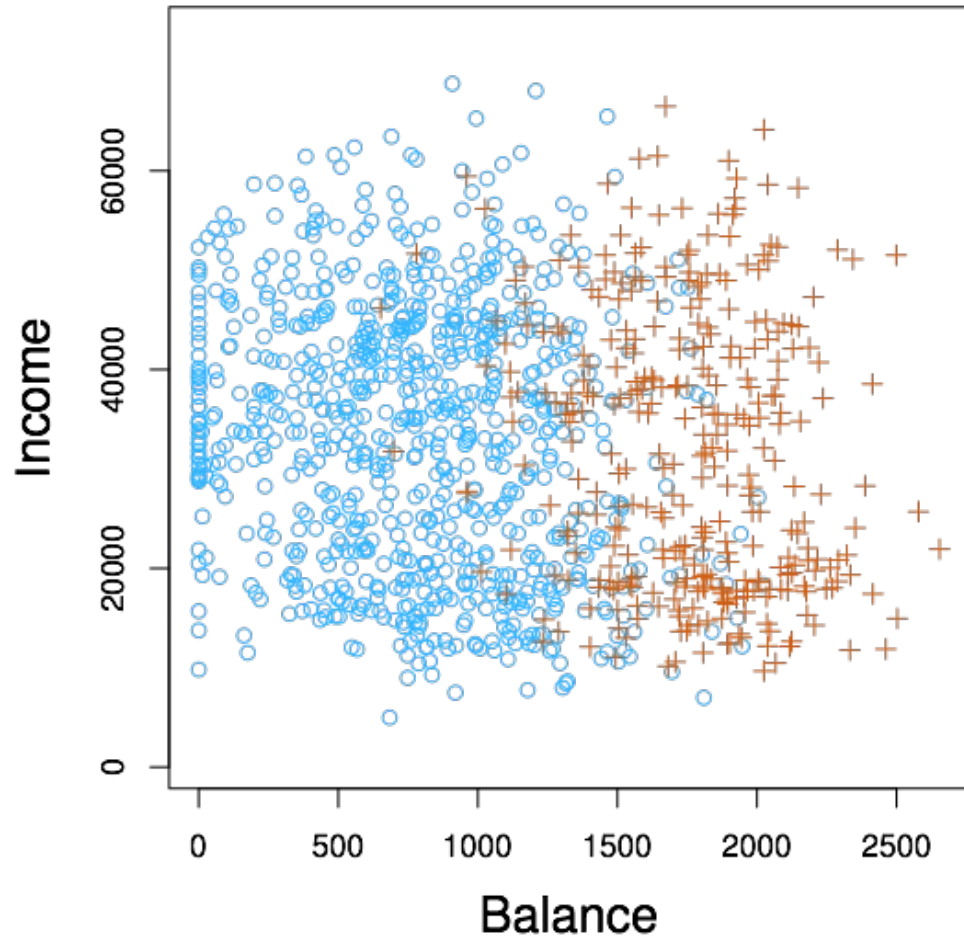
Y : default (1), not default (0)

$$p(X) = \mathbb{P}(Y = 1|X) \quad ?$$

$$p(X) = \mathbb{P}(Y = 1|X) > \tau$$



threshold



$$p(X) = \mathbb{P}(Y = 1|X) \quad ?$$

$$p = 1$$

$$p(X) = \beta_0 + \beta_1 X \longrightarrow (-\infty, +\infty)$$

$$p(X) = \sigma(\beta_0 + \beta_1 X) \longrightarrow [0, 1]$$

?

Sigmoid Function

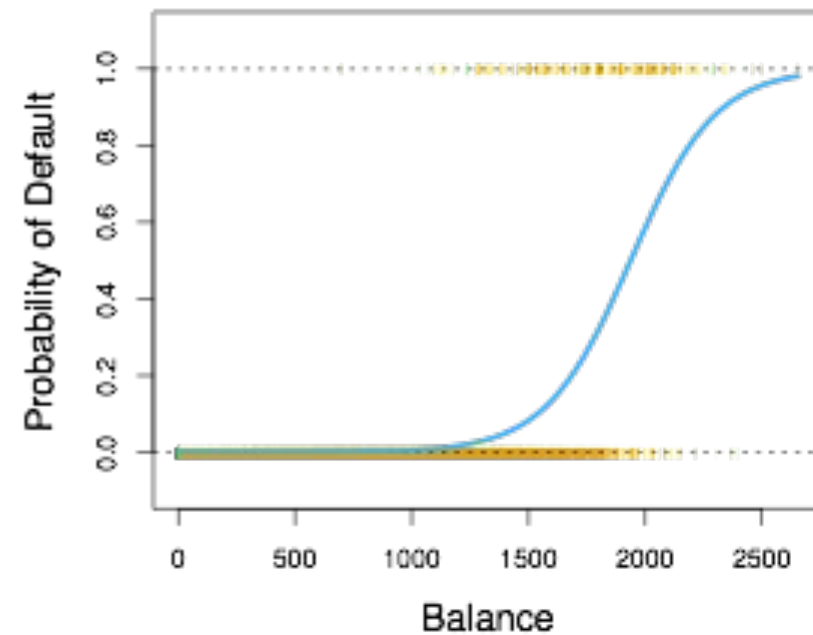
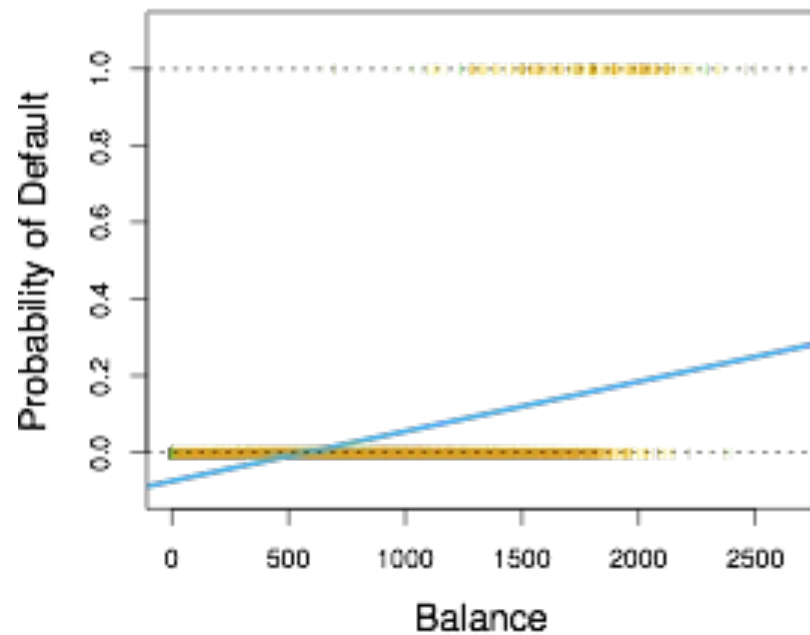
$$\sigma(y) = \frac{e^y}{1 + e^y}$$

Logistic Function

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \longrightarrow [0, 1]$$

$$p(X) = \mathbb{P}(Y = 1|X) \quad ?$$

$$p(X) = \beta_0 + \beta_1 X \longrightarrow (-\infty, +\infty) \quad p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \longrightarrow [0, 1]$$



$$\log \left(\underbrace{\frac{p(X)}{1 - p(X)}}_{\text{odds}} \right) = \beta_0 + \beta_1 X$$

$$p > 1$$

$$p(X) = \sigma(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)$$

Logistic Function

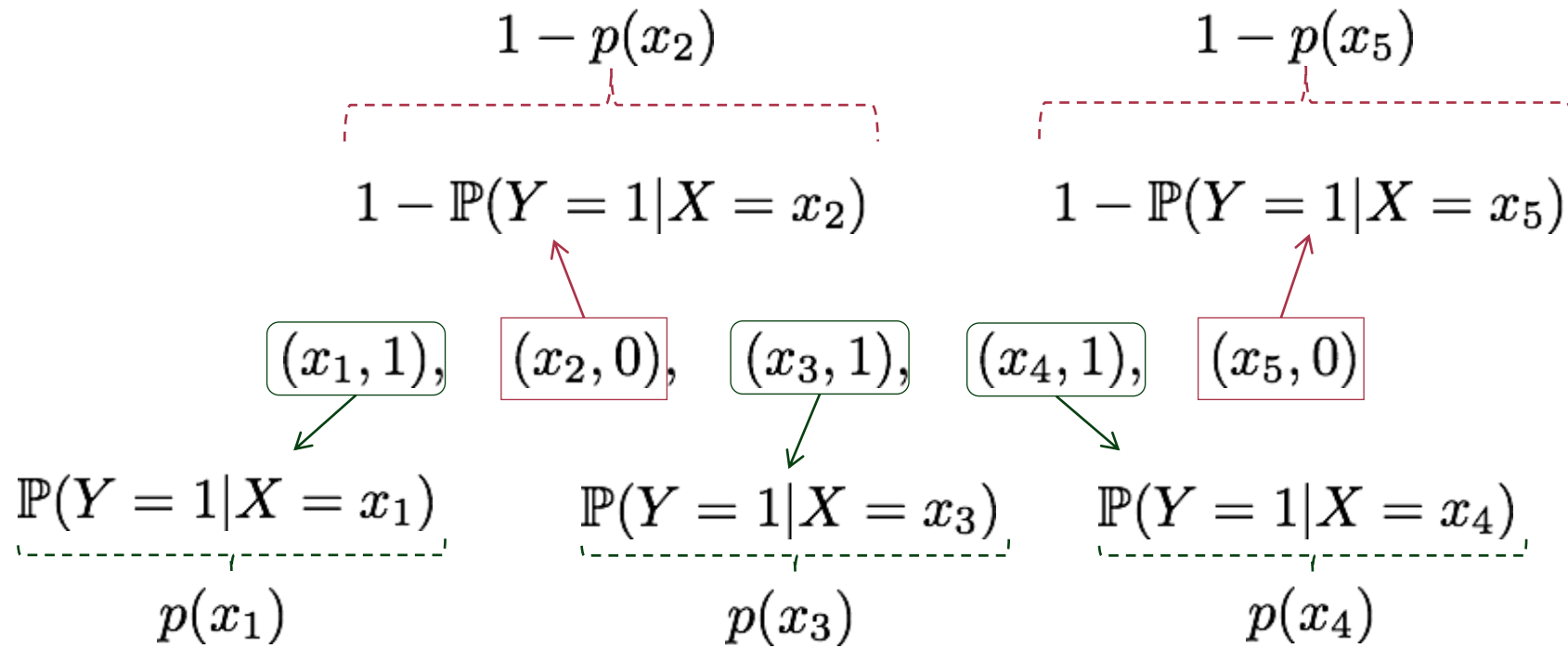
$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

$$\beta_0, \beta_1, \dots, \beta_p?$$



training data

$$\{(x_i, y_i) : 1, \dots, n\}$$



$$p(x_1)p(x_3)p(x_4)(1 - p(x_2))(1 - p(x_5))$$

Likelihood Function

$$\ell(\beta_0, \dots, \beta_p) = \prod_{i:y_i=1} p(x_i) \prod_{j:y_j=0} (1 - p(x_j))$$

Maximum Likelihood

$$\ell(\beta_0, \dots, \beta_p) = \prod_{i:y_i=1} p(x_i) \prod_{j:y_j=0} (1 - p(x_j))$$

$$\max_{\beta_0, \dots, \beta_p} \ell(\beta_0, \dots, \beta_p)$$



$$\max_{\beta_0, \dots, \beta_p} \log(\ell(\beta_0, \dots, \beta_p))$$



$$\max_{\beta_0, \dots, \beta_p} \sum_{i:y_i=1} p(x_i) + \sum_{j:y_j=0} (1 - p(x_j))$$



Concave Function



$$\hat{\beta}_0, \dots, \hat{\beta}_p$$

$$p = 1$$

	Coefficient	Std. Error	Z-statistics	<i>P-value</i>
Intercept	-3.5041	0.0707	-49.55	< 0.0001
Student (1)	0.4049	0.1150	3.52	0.0004

$$p > 1$$

	Coefficient	Std. Error	Z-statistics	<i>P-value</i>
Intercept	-10.8690	0.4923	-22.08	< 0.0001
Balance	0.0057	0.0002	24.74	< 0.0001
Income	0.0030	0.0082	0.37	0.7115
Student (1)	-0.6468	0.2362	-2.74	0.0062

Confounding Effect

Prediction

	Coefficient	Std. Error	Z-statistics	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
Balance	0.0057	0.0002	24.74	< 0.0001
Income	0.0030	0.0082	0.37	0.7115
Student (1)	-0.6468	0.2362	-2.74	0.0062

$$X = \begin{bmatrix} 1,500 \\ 40,000 \\ 1 \end{bmatrix}$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}$$

$$\hat{p}(X) = \frac{e^{-10.869 + 0.0057 \times 1,500 + 0.003 \times 40 - 0.6468 \times 1}}{1 + e^{-10.869 + 0.0057 \times 1,500 + 0.003 \times 40 - 0.6468 \times 1}} = 0.058$$

$$p(X) = \mathbb{P}(Y = 1|X) > \tau$$

$\tau = 0.5$		True		
		No	Yes	Total
Predicted	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

$\tau = 0.2$		True		
		No	Yes	Total
Predicted	No	9,432	138	9,570
	Yes	235	195	430
	Total	9,667	333	10,000

Confusion Matrix

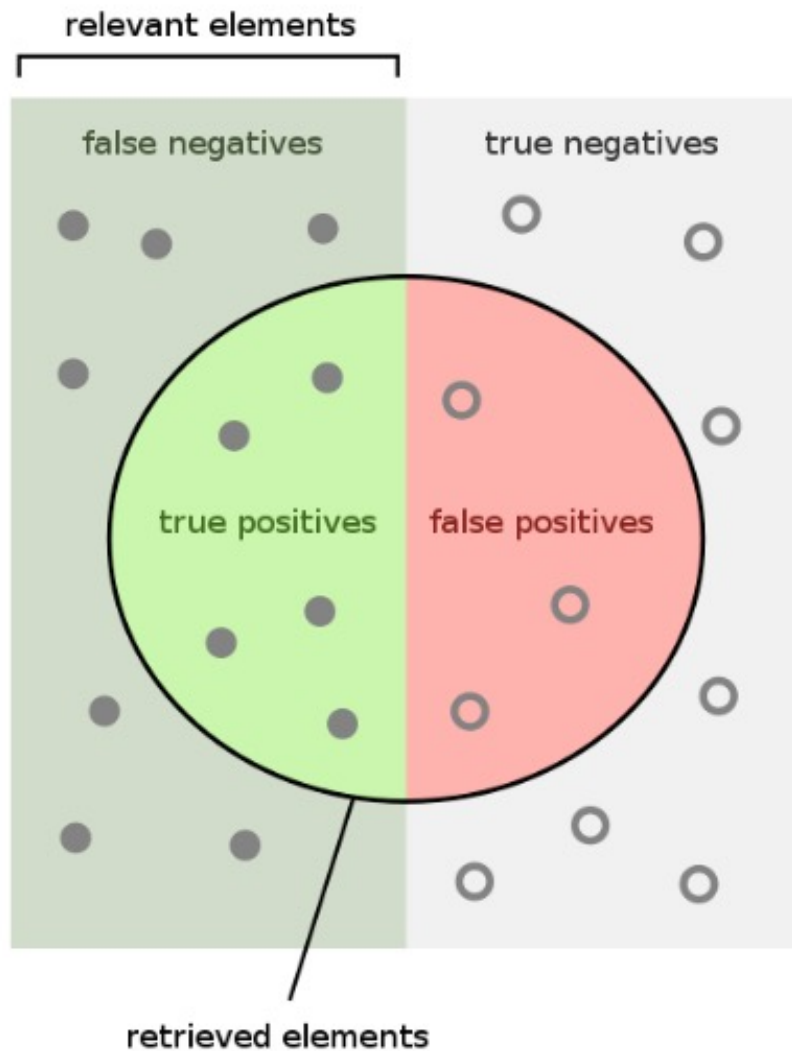
		Predicted Class		
		-	+	Total
True Class	-	True Negative (TN)	False Positive (FP)	N
	+	False Negative (FN)	True Positive (TP)	P
	Total	N*	P*	

Confusion Matrix

		Predicted Class		
		-	+	Total
True Class	-	TN	FP	N
	+	FN	TP	P
	Total	N*	P*	

Name	Defn.	Synonyms
False Positive Rate	FP/N	type I error, (1-specificity)
True Positive Rate	TP/P	(1-type II error), power, sensitivity, recall
Positive Pred. Value	TP/P*	precision, (1-false discovery proportion)
Negative Pred. Value	TN/N*	

Precision and Recall



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

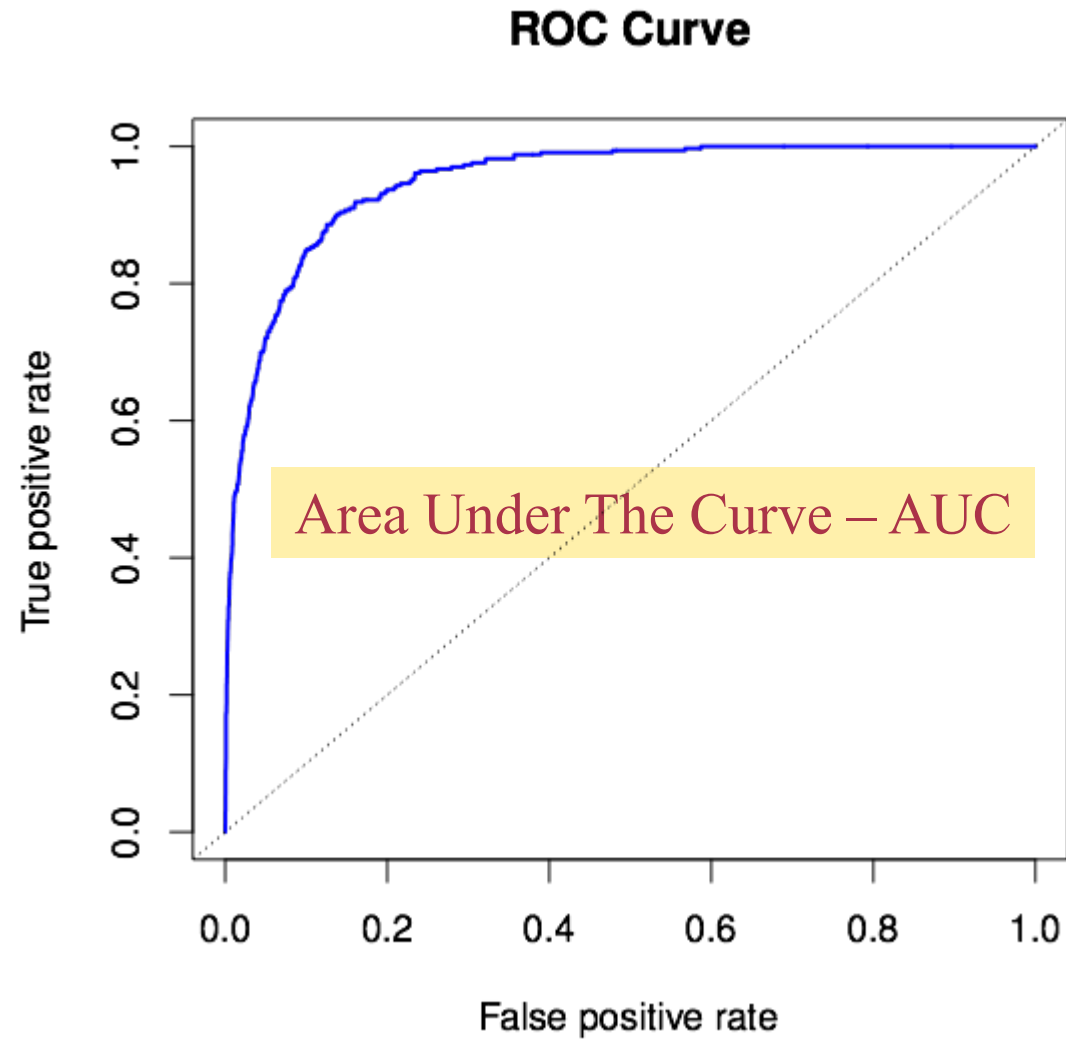
How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

*Wikipedia

ROC* Curve



* Receiver Operating Characteristics

Logistic Regression in Action

```
[1]: import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
```

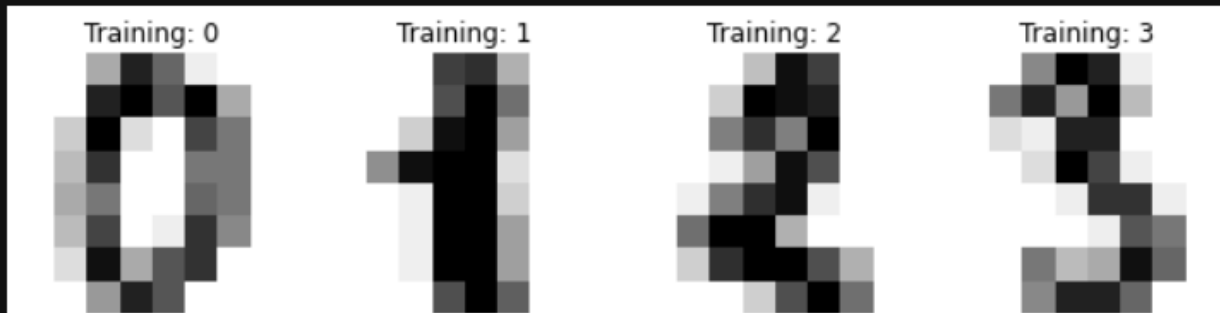
Adding packages and functions

```
[2]: digits = datasets.load_digits()

digits.keys()
```

Loading the dataset and checking the contents

```
[3]: _, axes = plt.subplots(nrows=1, ncols=4, figsize=(10, 3))
for ax, image, label in zip(axes, digits.images, digits.target):
    ax.set_axis_off()
    ax.imshow(image, cmap=plt.cm.gray_r, interpolation="nearest")
    ax.set_title("Training: %i" % label)
```



Displaying first 4 images with their labels


```
[4]: X = digits.data/digits.data.max()

value = 5
y = np.array([1 if target==value else 0 for target in digits.target])
```

Rescaling inputs and changing the output to binary classification

```
[5]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.5, shuffle=False
)
```

Splitting the dataset into 50% training and 50% test subsets

```
[6]: LR = LogisticRegression(random_state=42)
LR.fit(X_train, y_train)

y_pred = LR.predict(X_test)

accuracy_score(y_test, y_pred)
```

Fitting logistic regression model and checking the accuracy of the prediction

```
[6]: 0.9844271412680756
```

```
[7]: print(
    f"Classification report for classifier {LR}:\n\n"
    f"{classification_report(y_test, y_pred)}\n"
)
```

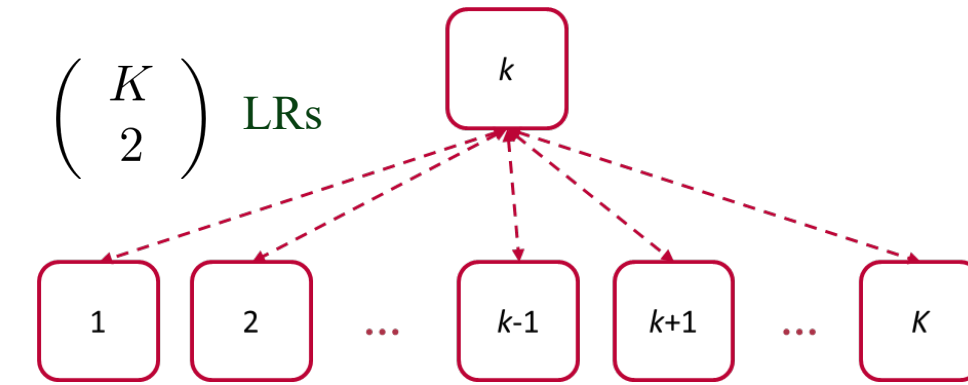
```
Classification report for classifier LogisticRegression(random_state=42):
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	808
1	0.94	0.90	0.92	91
accuracy			0.98	899

Checking the precision and recall

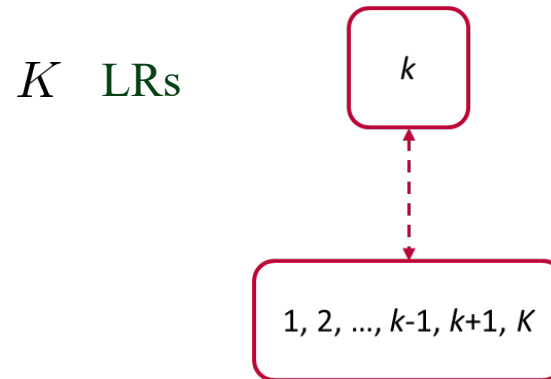
Multiclass Classification

One-Versus-One
(All-Pairs)



x_0 is assigned to the most frequently assigned class

One-Versus-Rest



x_0 is assigned
to the class
that attains the
highest value

```
[8]: from sklearn.multiclass import OneVsRestClassifier

X = digits.data/digits.data.max()
y = digits.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.5, shuffle=False
)

clf = OneVsRestClassifier(LogisticRegression(random_state=42))
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)

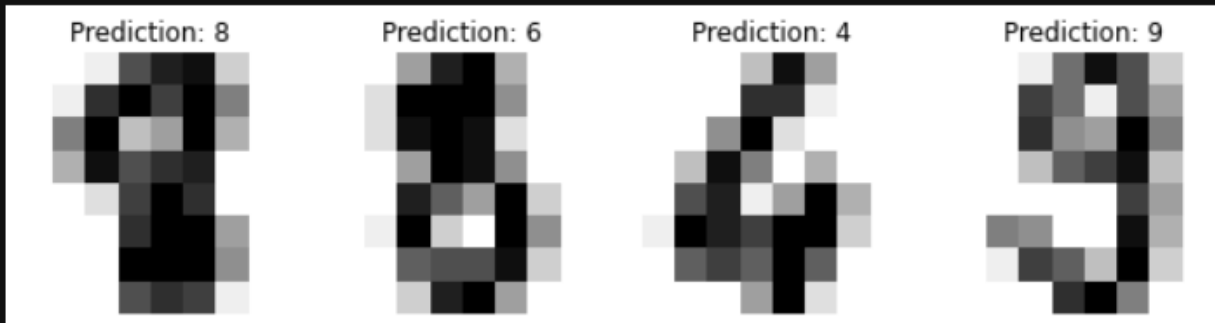
print(
    f"Classification report for classifier {clf}:\n"
    f"{classification_report(y_test, y_pred)}\n"
)
```

Classification report for classifier OneVsRestClassifier(estimator=LogisticRegression(random_state=42)):

	precision	recall	f1-score	support
0	0.96	0.99	0.97	88
1	0.90	0.85	0.87	91
2	1.00	0.98	0.99	86
3	0.97	0.85	0.91	91
4	0.98	0.93	0.96	92
5	0.90	0.95	0.92	91
6	0.96	0.99	0.97	91
7	0.96	0.99	0.97	89
8	0.93	0.88	0.90	88
9	0.81	0.93	0.87	92
accuracy			0.93	899

Using one-vs-rest method and the binary logistic regression classifier, we solve the multi-class problem

```
[9]: _, axes = plt.subplots(nrows=1, ncols=4, figsize=(10, 3))
for ax, image, prediction in zip(axes, X_test, y_pred):
    ax.set_axis_off()
    image = image.reshape(8, 8)
    ax.imshow(image, cmap=plt.cm.gray_r, interpolation="nearest")
    ax.set_title(f"Prediction: {prediction}")
```



Displaying the predictions